Customer Data Analytics and Visualization Dashboard Report

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Project Overview

This project explores a customer dataset containing 15 columns with different data types. Using Microsoft Excel, I examined the dataset through column filters, sorted annual income in descending order, and performed basic calculations to summarize key customer information.

To make the data easier to interpret, I created a dashboard featuring key customer metrics, a gender distribution doughnut chart, a bar chart showing customer distribution by city, and a line chart illustrating purchase history over time.

The dashboard provides a visual overview of customer demographics, income, subscription status, and purchasing activity, making the information easier to review and compare.

Objectives

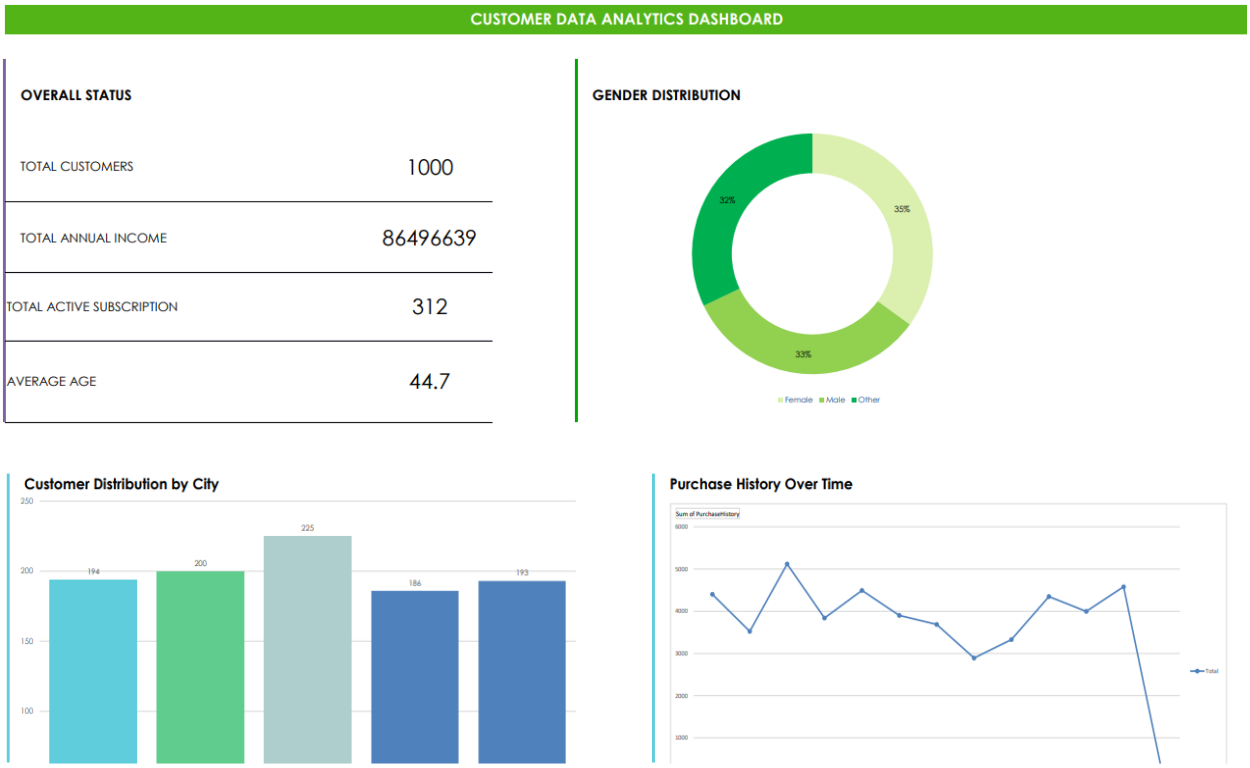
- To explore the dataset and summarize selected customer metrics through filtering, basic calculations, and visualizations.

Dataset Description

The dataset contains 1000 customer records and 15 columns (CustomerID, FirstName, LastName, Email, PhoneNumber, Age, Gender, Country, City, AnnualIncome, PurchaseHistory, LastPurchaseDate, SubsciptionStatus, FeedbackScore, LoyaltyPoints). The columns consists of different datatypes such as text, number, and date.

In exploring the dataset, I used filtering that includes sorting the annual income in descending order. I also did basic calculations used for summary metrics. In creating the dashboard, I used doughnut chart, bar graph and line graph to properly visualize the dataset and to see the relationship between datas.

Dashboard



Overall Status

The dashboard presents key customer metrics, including the total number of customers, total annual income, number of active subscriptions, and average customer age. These indicators provide a concise overview of the dataset and summarize essential information about the customer base.

Gender Distribution

The chart shows that female customers represent the largest proportion of the dataset at 35%, followed by male customers at

33% and customers categorized as “Other” at 32%. The distribution is relatively balanced across the three categories, with only a small difference in their respective proportions.

Customer Distribution by City

The bar chart shows that New York City has the highest number of customers, with a total of 225. It is followed by Mumbai with 200 customers, London with 194, Toronto with 193, and Sydney with 186. This indicates that New York City has the largest customer representation among the five cities, while Sydney has the smallest.

Purchase History over time

The line graph illustrates fluctuations in the total purchase history throughout 2023. During the first quarter (Q1), the recorded purchase value declined from January to February before increasing to its highest level of the quarter in March. In Q2, the values continued to fluctuate, while Q3 showed a decline in August followed by a gradual increase in September. In Q4, the purchase value rose in October, decreased in November, and reached its highest level of the quarter in December.

However, the graph shows a sharp decline in January 2024 compared with December 2023. This may indicate a decrease in recorded purchasing activity, although further investigation is needed to determine whether the change reflects actual customer behavior or incomplete data for the final month displayed.

Observations and Conclusion

The customer data analytics dashboard provides an overview of the customer base in terms of demographics, annual income, subscription status, geographic distribution, and purchase history. The gender distribution is relatively balanced, with female customers representing the largest category at 35%. Among the five cities presented, New York City has the highest customer count at 225, while Toronto has the lowest at 173.

The dashboard also summarizes key customer metrics, including an average age of 44.7 years and 312 active subscriptions out of 1,000 customers. Meanwhile, the purchase history chart shows fluctuations throughout 2023 and a sharp decrease in the displayed value for January 2024.

Overall, the project demonstrates how Microsoft Excel can be used to explore customer data and present important metrics and patterns through visualizations. Further analysis would be needed to understand the factors influencing purchasing activity, differences in customer behavior, and subscription status.